

MATH 231 – Linear Algebra I

Complex Numbers ■ and a First Look at Quaternions

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What you should be able to do afterwards. Explain why $x^2 + 1$ has no real roots but two complex ones; write a complex number in the form $a + bi$ and identify its real and imaginary parts; plot complex numbers on the complex plane; add, subtract, multiply and divide them; take a conjugate and say what it does geometrically; compute a modulus and use it to measure distance; describe a disc by an inequality; write a point of the unit circle as $\cos \theta + i \sin \theta$; state Euler's formula and Euler's identity; explain why multiplying by i is a rotation by 90° ; state de Moivre's formula and say what the roots of unity are; and describe what a quaternion is and where quaternions are used.

Further reading. Axler, *Linear Algebra Done Right*, Ch. 1 (the field $\mathbb{C}$) and Ch. 4 (polynomials, the fundamental theorem of algebra). Euler's formula, De Moivre, and quaternions are supplementary — see the optional appendices.

1. Where the real numbers run out

Start with a polynomial you can already handle. Let

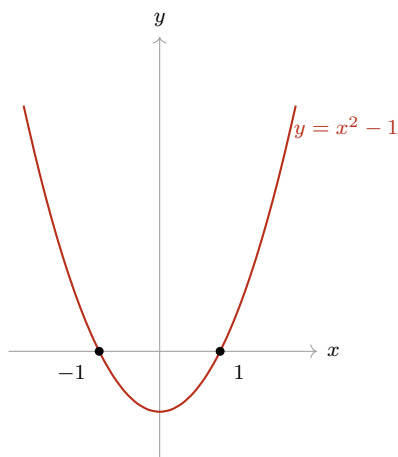
$$f(x) = x^2 - 1.$$

To find its roots, set $f(x) = 0$: then $x^2 = 1$, so $x = \pm 1$. Two distinct roots, both real, and the graph confirms them — the parabola crosses the x -axis at -1 and at $+1$.

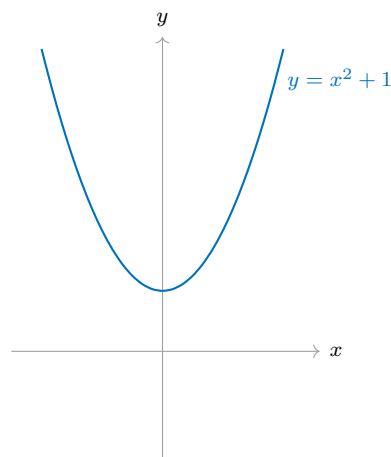
Now change one sign. Let

$$f(x) = x^2 + 1.$$

The same procedure gives $x^2 = -1$, so $x = \pm\sqrt{-1}$ — and here the procedure stops. There is no real number whose square is negative: squaring a positive number gives a positive number, squaring a negative number also gives a positive number, and $0^2 = 0$. So $\sqrt{-1}$ is undefined as a real quantity, and the graph agrees: the parabola $y = x^2 + 1$ never touches the x -axis at all.



two real roots



no real roots

Why the obstruction feels real

There is a reason $\sqrt{-1}$ looks not merely difficult but *meaningless*. Think about what a square root measures. If a square has area 25 square feet, then $\sqrt{25} = 5$ is the length of its side — a positive quantity, because lengths are positive. Ask for $\sqrt{-1}$ and you are asking for the side length of a square whose area is -1 square foot. No such square exists: area cannot be negative. The question has no answer because the question is malformed.

That was more or less the view for a long time. What changed was the discovery that if you *proceed anyway* — carry the impossible quantity through the computation and see what happens — the answers that come out at the end are correct, useful, and often real.

Historical note. Square roots of negative numbers first forced themselves on **Gerolamo Cardano** (Italian, 1501–1576) in his *Ars Magna* of **1545**, where they turned up while he was solving cubic equations. Cardano did the arithmetic but distrusted it, calling such quantities useless. It was **Rafael Bombelli** (Italian, 1526–1572) in his *L'Algebra* of **1572** who laid down consistent rules for calculating with them and showed that they could be manipulated safely — turning a curiosity into a tool. The notation i we use today was introduced much later, by **Leonhard Euler** (Swiss, 1707–1783) in **1777**.

2. The imaginary unit, and complex numbers

Definition 1 (The imaginary unit). The **imaginary unit** i is the number defined by the property

$$i^2 = -1.$$

It is informally written $i = \sqrt{-1}$, and read “the square root of minus one.”

Remark. Take $i^2 = -1$ as the definition and treat $i = \sqrt{-1}$ as shorthand for it. The radical notation is convenient but it invites a false move: the familiar rule $\sqrt{a}\sqrt{b} = \sqrt{ab}$ is a theorem about *non-negative* reals, and applying it here would give the nonsense $\sqrt{-1}\sqrt{-1} = \sqrt{1} = 1$ instead of the correct $i^2 = -1$. Work from $i^2 = -1$ and you will never trip over this.

With i available, the polynomial that defeated us is solved. Since $i^2 = -1$ and $(-i)^2 = (-1)^2 i^2 = i^2 = -1$, both i and $-i$ satisfy $x^2 = -1$. So

$$f(x) = x^2 + 1 \quad \text{has roots} \quad x = \pm i.$$

The graph told us the truth: f has no *real* roots. What it could not show us is that f has two *imaginary* roots, because they do not live on the x -axis. They live on a different line, in a plane we have not drawn yet.

Definition 2 (Complex number). A **complex number** is a number of the form

$$z = a + bi, \quad a, b \in \mathbb{R},$$

and the set of all of them is denoted $\mathbb{C}$:

$$\mathbb{C} = \{ a + bi : a, b \in \mathbb{R} \}.$$

We write $z \in \mathbb{C}$. The real number a is the **real part** of z and the real number b is the **imaginary part**:

$$\operatorname{Re}(z) := a, \quad \operatorname{Im}(z) := b.$$

Remark. The imaginary part is the real number b , *not* bi . For $z = 3 + 2i$ we have $\operatorname{Re}(z) = 3$ and $\operatorname{Im}(z) = 2$ — both plain real numbers. Two complex numbers are equal exactly when their real parts agree and their imaginary parts agree.

3. The complex plane

A complex number carries two independent real numbers, so it needs two axes to be drawn. Geometrically, complex numbers are **points of a plane** — the **complex plane** — laid out as follows:

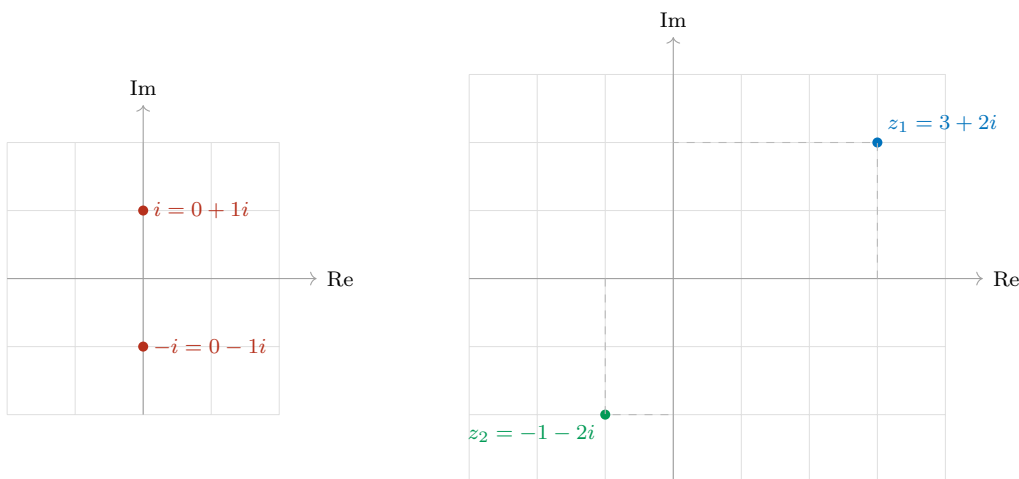
- the horizontal axis carries the *real* part, and is called the **real axis**;
- the vertical axis carries the *imaginary* part, and is called the **imaginary axis**.

So $z = a + bi$ is plotted at the point (a, b) .

Start with the two roots we just found. Both i and $-i$ have real part 0: writing them out in full, $i = 0 + 1i$ and $-i = 0 + (-1)i$, so

$$\operatorname{Re}(i) = 0, \quad \operatorname{Im}(i) = 1, \quad \operatorname{Re}(-i) = 0, \quad \operatorname{Im}(-i) = -1.$$

A real part of 0 means no horizontal displacement, so both numbers sit *on* the imaginary axis — one unit above the origin and one unit below. This is the sense in which the roots of $x^2 + 1$ were invisible on our earlier graph: they were never on the horizontal axis to begin with.



In the right-hand picture the dashed lines read off the two parts: $z_1 = 3 + 2i$ sits 3 to the right and 2 up, while $z_2 = -1 - 2i$ sits 1 to the left and 2 down.

3.1. Every real number is a complex number

What is a complex number with imaginary part 0? It is $z = a + 0i = a$ — an ordinary real number, sitting on the real axis. So the real numbers are not a separate species; they are exactly the complex numbers whose imaginary part vanishes. Every real number is a complex number, which is to say

$$\mathbb{R} \subset \mathbb{C}.$$

The real line is the horizontal axis of the complex plane, and $\mathbb{C}$ is what you get by allowing motion off that line.

3.2. Complex numbers are two-dimensional

The real numbers form a line: one number pins down a position, so $\mathbb{R}$ is one-dimensional. A complex number needs *two* real numbers, chosen independently, to pin down its position in the plane — two degrees of freedom. So $\mathbb{C}$ is two-dimensional.

Connection. This is the same counting we used for $\mathbb{R}^2$ in the vectors lecture, and the resemblance is not a coincidence. A complex number $a + bi$ and a vector $[a, b]$ carry identical data. Accordingly, several operations on complex numbers are performed *componentwise*, exactly as with vectors, and for those operations the two objects are genuinely interchangeable.

Looking ahead. Where they stop being interchangeable is multiplication. There is no useful way to multiply two vectors in $\mathbb{R}^2$ and get a third vector — but complex numbers multiply, and divide, perfectly well. That extra structure is the whole reason $\mathbb{C}$ deserves its own name, and it is what makes the rotation results of §7 possible.

4. Arithmetic

4.1. Addition and subtraction

Collect real parts with real parts and imaginary parts with imaginary parts.

Definition 3 (Addition and subtraction in $\mathbb{C}$). For $z = a + bi$ and $w = c + di$ in $\mathbb{C}$,

$$z + w = (a + c) + (b + d)i, \quad z - w = (a - c) + (b - d)i.$$

Both are componentwise, and both return a complex number — so $\mathbb{C}$ is closed under addition and subtraction.

Example 1. $(3 + 2i) + (1 + 4i) = (3 + 1) + (2 + 4)i = 4 + 6i$.

Example 2. $(5 - 3i) - (2 + i) = (5 - 2) + (-3 - 1)i = 3 - 4i$.

Example 3. $(-2 + 5i) + (6 - 5i) = (-2 + 6) + (5 - 5)i = 4 + 0i = 4$.

Worth noticing: the sum of two numbers that are *not* real can be real. The imaginary parts cancelled.

4.2. Multiplication

Multiplication needs no new rule at all. Expand the product as you would any two binomials, and then use $i^2 = -1$ to clean up:

$$(a + bi)(c + di) = ac + adi + bci + bd i^2 = ac + adi + bci - bd,$$

and collecting the parts,

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i$$

The single step that does all the work is $bd i^2 = -bd$: a term that started out imaginary lands in the real part, with its sign flipped.

Example 4. $(3 + 2i)(1 + 4i) = 3 + 12i + 2i + 8i^2 = 3 + 14i - 8 = -5 + 14i$.

Example 5. $(2 + 3i)(2 - 3i) = 4 - 6i + 6i - 9i^2 = 4 + 9 = 13$.

The imaginary parts cancelled and i^2 turned $-9i^2$ into $+9$, leaving a real answer. This is not an accident, and §4.3 explains it.

Remark. Multiplication is emphatically *not* componentwise: the real part of the product mixes a , b , c and d together. Compare addition, where the first slots never talk to the second. This is precisely the point at which a complex number stops behaving like a vector in $\mathbb{R}^2$.

4.3. The conjugate

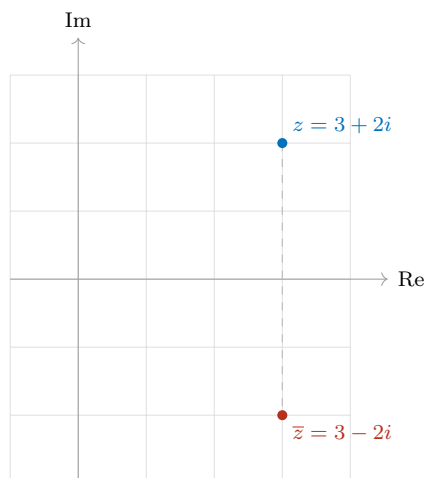
Definition 4 (Complex conjugate). The **conjugate** of $z = a + bi$ is

$$\bar{z} = a - bi,$$

obtained by flipping the sign of the imaginary part. It is read “ z -bar.”

Remark. Some texts write z^* for the conjugate. We use the overline $\bar{z}$ throughout, and we specifically avoid the hat: in the vectors lecture $\hat{\mathbf{u}}$ already means the *normalization* of $\mathbf{u}$, and reusing the hat here would collide.

Geometrically, conjugation is a **reflection about the real axis**. The real part is untouched, so the horizontal position stays; the imaginary part changes sign, so the point flips to the other side of the horizontal axis. It is the mirror image of z in the real axis.



Two facts about conjugation that we will use immediately. First, a number equals its own conjugate exactly when $b = 0$, that is, exactly when it is real. Second, and this is the useful one:

$$z\bar{z} = (a + bi)(a - bi) = a^2 - abi + abi - b^2i^2 = a^2 + b^2.$$

So $z\bar{z}$ is always a *real*, non-negative number — the imaginary parts cancel by construction. That is what happened in the second multiplication example above, where $(2 + 3i)(2 - 3i) = 13$.

4.4. Division

Division is possible in $\mathbb{C}$ too, provided the denominator is not zero. The obstacle is cosmetic: a quotient like $(3 + 2i)/(1 - i)$ is a perfectly good complex number, but it is not yet written in the form $a + bi$, so we cannot read off its real and imaginary parts.

The trick is the previous subsection. We want a *real* denominator, and $w\bar{w}$ is always real — so multiply the numerator and the denominator by the conjugate of the denominator.

To divide. Let $z = a + bi$ and $w = c + di$ with $w \neq 0$ (that is, c and d not both zero). Then

$$\frac{z}{w} = \frac{z}{w} \cdot \frac{\bar{w}}{\bar{w}} = \frac{(a + bi)(c - di)}{(c + di)(c - di)} = \frac{(ac + bd) + (bc - ad)i}{c^2 + d^2} = \frac{ac + bd}{c^2 + d^2} + \frac{bc - ad}{c^2 + d^2}i.$$

Since $w \neq 0$ we have $c^2 + d^2 > 0$, so the two fractions are honest real numbers and the result is in the form $a + bi$. Note that multiplying by $\bar{w}/\bar{w}$ multiplies by 1: the value is unchanged, only its appearance.

Example 6 (A division, worked out). Compute $\frac{3 + 2i}{1 - i}$.

The denominator is $w = 1 - i$, so $\bar{w} = 1 + i$. Multiply top and bottom by it:

$$\frac{3 + 2i}{1 - i} = \frac{3 + 2i}{1 - i} \cdot \frac{1 + i}{1 + i} = \frac{(3 + 2i)(1 + i)}{(1 - i)(1 + i)}.$$

Numerator: $(3 + 2i)(1 + i) = 3 + 3i + 2i + 2i^2 = 3 + 5i - 2 = 1 + 5i$.

Denominator: $(1 - i)(1 + i) = 1 + i - i - i^2 = 1 + 1 = 2$.

Therefore

$$\frac{3 + 2i}{1 - i} = \frac{1 + 5i}{2} = \frac{1}{2} + \frac{5}{2}i,$$

which is in the form $a + bi$ with $a = \frac{1}{2}$ and $b = \frac{5}{2}$.

Remark. So $\mathbb{C}$ supports addition, subtraction, multiplication, and division by anything nonzero. A number system with all four is called a *field*, and $\mathbb{C}$ is one — just as $\mathbb{R}$ is. This is why complex scalars can be used anywhere real scalars can, a fact we will lean on when eigenvalues turn out to be complex.

5. The modulus, distance, and discs

5.1. From absolute value to modulus

For a real number x , the absolute value $|x|$ measures the distance from x to 0, and $|x - a|$ measures the distance from x to a . That works because the real numbers lie on a *line*: there is only one direction to travel, so a single subtraction and a sign-strip is enough.

Complex numbers live on a plane, so distance needs the two-dimensional measurement instead.

Definition 5 (Modulus). The **modulus** of $z = a + bi$ is

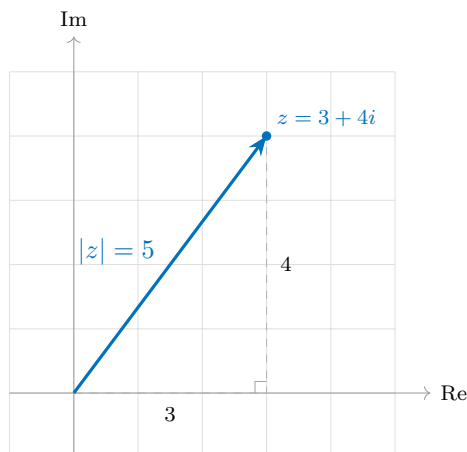
$$|z| = \sqrt{a^2 + b^2},$$

the Euclidean distance from z to the origin of the complex plane.

Connection. This is the norm of the vector $[a, b]$ from the vectors lecture, under a different name and the same Pythagoras: the horizontal leg has length $|a|$, the vertical leg $|b|$, and $|z|$ is the hypotenuse. The identity $z\bar{z} = a^2 + b^2$ of §4.3 can therefore be restated as

$$|z|^2 = z\bar{z},$$

which is the exact counterpart of $\langle \mathbf{u}, \mathbf{u} \rangle = \|\mathbf{u}\|^2$. Conjugation plays the role that the second copy of the vector played in the dot product.



Example 7. $|3 + 4i| = \sqrt{9 + 16} = \sqrt{25} = 5$.

Example 8. $|-1 + i| = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$.

Example 9. $|i| = \sqrt{0^2 + 1^2} = 1$, and likewise $|-i| = 1$. The two roots of $x^2 + 1$ both sit at distance 1 from the origin.

5.2. Distance between two complex numbers

The modulus also measures distances between points, exactly as absolute value does on the line: for $z, w \in \mathbb{C}$,

$$|z - w| = \text{the distance from } z \text{ to } w.$$

Subtract to get the displacement between them, then take its modulus to get the displacement's length. The order does not matter, since $|z - w| = |w - z|$.

Example 10. The distance from $z = 3 + 2i$ to $w = 1 - i$ is

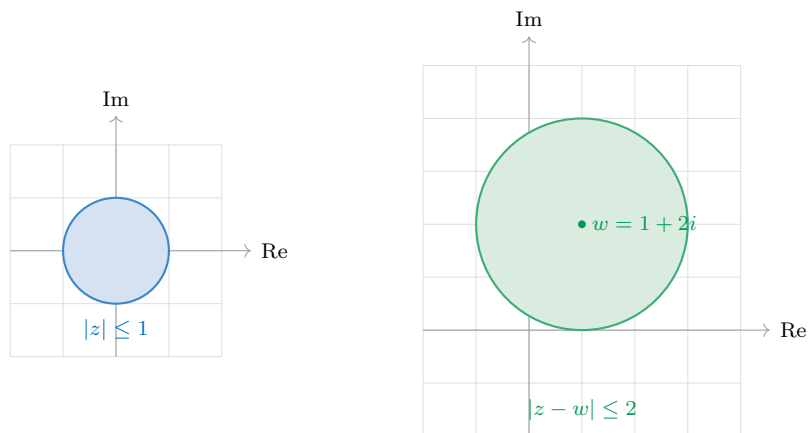
$$|z - w| = |(3 - 1) + (2 - (-1))i| = |2 + 3i| = \sqrt{4 + 9} = \sqrt{13}.$$

5.3. Discs

Once distance is available, whole *regions* of the complex plane can be described by a single inequality. The most common such regions in mathematics are **discs**.

- $|z| \leq 1$ is the set of complex numbers at distance at most 1 from the origin: the **disc of radius 1 centred at the origin**, boundary included. It is called the *closed unit disc*.
- $|z - w| \leq 2$ with $w = 1 + 2i$ is the set of complex numbers at distance at most 2 from the point $1 + 2i$: the **disc of radius 2 centred at $1 + 2i$** .

The pattern generalises: $|z - w| \leq r$ is the disc of radius r centred at w . Replacing $\leq$ by $<$ removes the boundary circle, and replacing it by $=$ leaves the boundary circle alone.



6. The unit circle and Euler's formula

Narrow attention to the complex numbers of modulus exactly 1:

$$|z| = 1.$$

By the previous section this is a circle of radius 1 centred at the origin — the **unit circle** of the complex plane.

Connection. This is the same set, drawn in the same place, as the unit vectors $\mathbb{S}^1$ of the vectors lecture: $|z| = 1$ is $a^2 + b^2 = 1$, which is what $\|\mathbf{u}\| = 1$ was. The complex numbers of unit modulus *are* the unit vectors, now equipped with the ability to multiply one another.

6.1. Trigonometric form

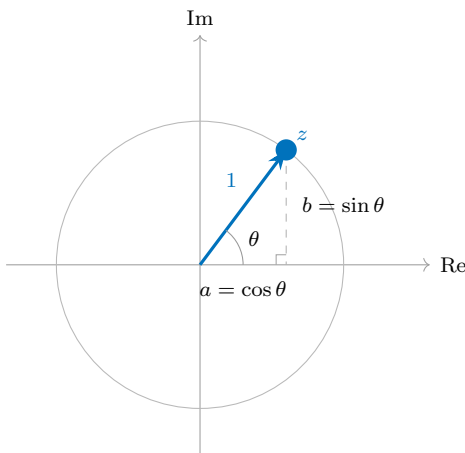
Take a point $z = a + bi$ on the unit circle and drop a perpendicular from it to the real axis. This produces a right triangle whose hypotenuse runs from the origin to z and therefore has length $|z| = 1$. Call θ the angle at the origin, measured from the positive real axis up to the hypotenuse.

Now read off the two legs by trigonometry. The horizontal leg is adjacent to θ and the vertical leg is opposite it, and the hypotenuse is 1, so

$$\cos \theta = \frac{a}{1} = a, \quad \sin \theta = \frac{b}{1} = b.$$

The two parts of z are the cosine and the sine:

$$z = \cos \theta + i \sin \theta$$



Notice what has happened to the number of variables. A general complex number needed two real numbers. A complex number *on the unit circle* needs only one, namely θ : the modulus is already fixed at 1, so only the angle is free. So z has become a function of θ , and varying θ walks z around the circle — once around for every increase of 2π .

6.2. Euler's formula

That combination of cosine and sine has a startling second description.

Euler's formula. For every real θ ,

$$e^{i\theta} = \cos \theta + i \sin \theta,$$

where $e^x = \exp(x)$ is the ordinary exponential function.

Understanding *why* an exponential should equal a combination of trigonometric functions is beyond what we need today. If you would like to see it, Appendix A derives the formula from scratch using the Maclaurin series of e^x , $\cos x$ and $\sin x$; the derivation is short and genuinely one of the most satisfying calculations in mathematics.

For now, take Euler's formula as a licence to write any point of the unit circle as a single exponential. It immediately yields the most quoted equation in the subject. Put $\theta = \pi$:

$$e^{i\pi} = \cos \pi + i \sin \pi = -1 + 0i = -1,$$

which is usually written

Euler's identity.

$$e^{i\pi} + 1 = 0.$$

Five of the most important constants in mathematics — e , i , π , 1 and 0 — tied together by one equation containing nothing else.

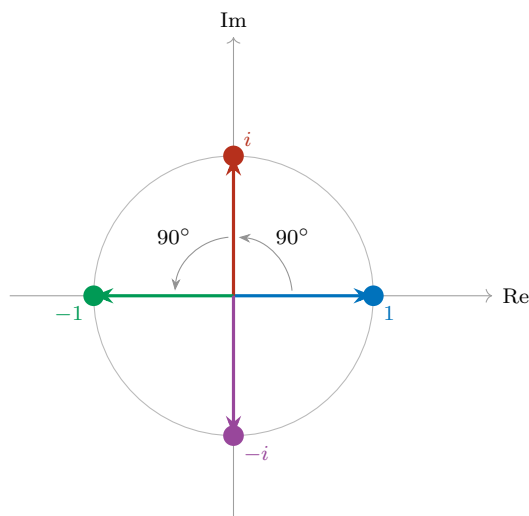
7. Multiplying by i is a rotation

Here is the property that makes complex numbers indispensable in applications. We will find it first by experiment and then prove it.

Start with $z = 1$. Written in full that is $1 + 0i$, so $a = 1$ and $b = 0$, and $|z| = \sqrt{1+0} = 1$ — it is on the unit circle. Now multiply repeatedly by i and watch where the point goes:

$$1 \xrightarrow{\times i} i \xrightarrow{\times i} i^2 = -1 \xrightarrow{\times i} -i \xrightarrow{\times i} -i^2 = 1.$$

Each step turns the point through a quarter turn counter-clockwise, and four steps return it to where it began.



Proving the angle is exactly 90°

Experiment is not proof, and we have the tools to settle it. Take an arbitrary complex number $z = a + bi$ and multiply by i :

$$w = (a + bi)i = ai + bi^2 = -b + ai.$$

Now hand both numbers to the vectors lecture. As vectors,

$$z \longleftrightarrow \mathbf{z} = [a, b], \quad w \longleftrightarrow \mathbf{w} = [-b, a],$$

and their dot product is

$$\langle \mathbf{z}, \mathbf{w} \rangle = a(-b) + b(a) = -ab + ab = 0.$$

By the orthogonality test, a dot product of zero means the two are orthogonal — so z and iz meet at a right angle, for *every* z . That is the 90° claim, proved in three lines.

One detail finishes the job. Orthogonality alone would permit the length to change as well; but

$$|w| = \sqrt{(-b)^2 + a^2} = \sqrt{a^2 + b^2} = |z|,$$

so multiplication by i preserves the modulus. Same distance from the origin, turned through a right angle: that is a *rotation*, not a rotation combined with a stretch.

Multiplication by i rotates a complex number by 90° **counter-clockwise** about the origin.

And the mirror statement holds too. Multiplying by $-i$ gives $(a + bi)(-i) = -ai - bi^2 = b - ai$, i.e. the vector $[b, -a]$, whose dot product with $[a, b]$ is $ab - ba = 0$ — again orthogonal, again the same modulus, but the turn goes the other way:

Multiplication by $-i$ rotates a complex number by 90° **clockwise** about the origin.

Looking ahead. A quarter turn is the special case $\theta = \pi/2$ of something much more general. Since $i = e^{i\pi/2}$ by Euler’s formula, multiplying by i is multiplying by $e^{i\pi/2}$; and multiplying by $e^{i\theta}$ turns out to rotate by θ , for any angle at all. §8 puts that to work.

8. De Moivre’s formula, the roots of unity, and how many roots there are

8.1. De Moivre’s formula

Return to Euler’s formula, and rename the variable x to emphasise that it need not be thought of as an angle on a particular picture:

$$e^{ix} = \cos x + i \sin x.$$

This function is sometimes written $\text{cis}(x)$ — “cosine plus i sine” — which is a useful abbreviation to recognise.

Now raise both sides to an integer power n :

$$(e^{ix})^n = (\cos x + i \sin x)^n.$$

The left side simplifies by the ordinary law of exponents, $(e^{ix})^n = e^{inx}$. And e^{inx} is just Euler’s formula again, read with xn in place of x :

$$e^{i(xn)} = \cos(xn) + i \sin(xn).$$

Comparing the two expressions for the same quantity gives:

De Moivre’s formula. For every real x and every integer n ,

$$(\cos x + i \sin x)^n = \cos(nx) + i \sin(nx).$$

Read it as a statement about work saved. The left-hand side asks you to multiply out an n -th power of a binomial; the right-hand side says the answer is obtained by simply *multiplying the angle by n* . Raising to a power on the unit circle is nothing more than scaling an angle.

8.2. The roots of unity

De Moivre’s formula makes short work of a classical question.

The **n -th roots of unity** are the complex numbers that give 1 when raised to the n -th power — the solutions of $z^n = 1$.

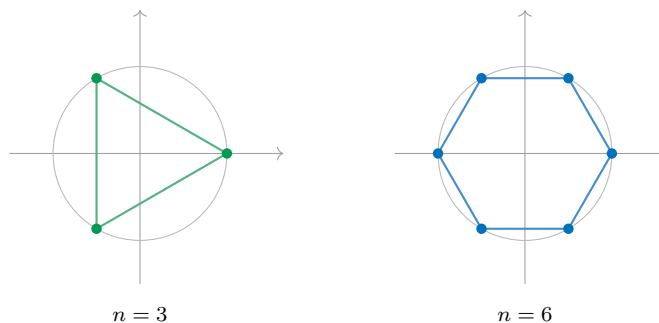
Over the real numbers this question is dull: $z^2 = 1$ has the two solutions ± 1 , and $z^3 = 1$ has only $z = 1$. Over the complex numbers it is beautiful. There are always exactly n solutions, and de Moivre says where: taking $z = \cos \theta + i \sin \theta$ on the unit circle, $z^n = 1$ requires $\cos(n\theta) + i \sin(n\theta) = 1$,

which happens exactly when $n\theta$ is a whole number of full turns. So $\theta = 2\pi k/n$ for $k = 0, 1, \dots, n-1$, giving

$$z_k = \cos \frac{2\pi k}{n} + i \sin \frac{2\pi k}{n} = e^{2\pi i k/n}.$$

Geometrically the answer is as clean as it could be: the n roots are spaced evenly around the unit circle, one of them always at 1, and joining them up produces a regular n -gon.

Example 11 (The fourth roots of unity). For $n = 4$ the roots are 1, i , -1 , $-i$ — exactly the four points we cycled through in §6 by repeatedly multiplying by i . They form a square. Indeed $i^4 = (i^2)^2 = (-1)^2 = 1$, as required.



8.3. How many roots does a polynomial have?

We opened this lecture with a polynomial that had two real roots and one that had none. Over $\mathbb{C}$ that irregularity disappears entirely.

Fundamental theorem of algebra (also called *d'Alembert's theorem*). Every non-constant polynomial with complex coefficients has at least one complex root. Consequently a polynomial of degree n has exactly n complex roots, counted with multiplicity.

In plain terms: once you allow complex numbers, *every* polynomial factors completely, and the number of roots is simply the degree — no exceptions, no special cases. Our two opening examples both have degree 2 and therefore both have exactly two roots. For $x^2 - 1$ they are ± 1 and happen to be real; for $x^2 + 1$ they are $\pm i$ and are not. Nothing was ever missing from $x^2 + 1$; we were simply looking in a space that was too small.

Historical note. The result is named for **Jean-Baptiste le Rond d'Alembert**, who attempted a proof in 1746, though the first substantially complete proof is due to **Carl Friedrich Gauss** in 1799. Despite the name, every known proof uses tools from analysis or topology rather than algebra alone.

9. Where this gets used

The rotation property of §6 is not a curiosity; it is the reason complex numbers appear in engineering and computer graphics.

Rotating things in two dimensions

Suppose you want to rotate a point of the plane through an angle θ about the origin — a sprite in a game, an image in an editor, a robot's heading. Encode the point (x, y) as the complex number

$z = x + iy$ and multiply by $e^{i\theta}$:

$$ze^{i\theta} = (x + iy)(\cos \theta + i \sin \theta) = (x \cos \theta - y \sin \theta) + i(x \sin \theta + y \cos \theta).$$

Reading off the two parts, the rotated point is $(x \cos \theta - y \sin \theta, x \sin \theta + y \cos \theta)$. A rotation of the plane has become a *single multiplication*. Composing two rotations is then just $e^{i\alpha}e^{i\beta} = e^{i(\alpha+\beta)}$ — angles add, which is exactly what rotations should do.

Signals and the frequency domain

Alternating currents, audio, radio, and vibration are all described by sinusoids, and a sinusoid is far easier to handle as a single complex exponential $e^{i\omega t}$ than as a sum of a sine and a cosine. Electrical engineers call such a complex number a *phasor*. The discrete Fourier transform — and the fast Fourier transform that computes it, one of the most heavily used algorithms in existence — is built entirely out of the roots of unity from §7.2.

Iteration in the plane

Some of the most familiar images in mathematics come from iterating a very simple complex function. Repeatedly applying $z \mapsto z^2 + c$ and asking which starting values stay bounded produces the Mandelbrot set. The same iteration written with real coordinates is a two-variable mess; written with complex multiplication it is four characters long.

Looking ahead. Complex numbers will return in this course for a structural reason rather than a geometric one. When we compute eigenvalues, we solve a polynomial equation — and by §7.3 those solutions are guaranteed to exist only if we allow them to be complex. Rotation matrices, in particular, have no real eigenvalues at all, which is exactly what you would expect of a transformation that leaves no direction pointing where it started.

10. Quaternions

Complex numbers extended the reals by adding one new unit, i , and doubled the dimension from 1 to 2. It is natural to ask whether the trick repeats. It does — but not in three dimensions, and not without a cost.

Historical note. **William Rowan Hamilton** (Irish, 1805–1865) spent years trying to build a three-dimensional analogue of the complex numbers, and failed, because no such system exists. The answer came to him on **Monday, 16 October 1843**, while walking with his wife Helen along the Royal Canal in Dublin on his way to a meeting of the Royal Irish Academy. He realised that *four* dimensions were needed, and that the price was giving up commutativity. Unable to wait, he stopped at Broome Bridge (also written Brougham Bridge) and carved the defining relation into the stone with a pocket knife:

$$i^2 = j^2 = k^2 = ijk = -1.$$

A plaque on the bridge marks the spot.

10.1. What a quaternion is

Definition 6 (Quaternion). A **quaternion** is an object of the form

$$q = a + bi + cj + dk, \quad a, b, c, d \in \mathbb{R},$$

where i, j, k are called the **basis elements**. The set of all quaternions is denoted $\mathbb{H}$, after Hamilton, and we write $q \in \mathbb{H}$.

So a quaternion carries *four* real numbers where a complex number carried two. The real number a is called the **scalar part** of q ; the remainder $bi + cj + dk$ is the **vector part**. Setting $c = d = 0$ recovers a complex number, and setting $b = c = d = 0$ recovers a real number, so

$$\mathbb{R} \subset \mathbb{C} \subset \mathbb{H}.$$

10.2. How the basis elements multiply

Everything about quaternion arithmetic follows from Hamilton's carving. Each basis element squares to -1 , exactly as i does in $\mathbb{C}$:

$$i^2 = j^2 = k^2 = -1.$$

And the products of *distinct* basis elements cycle through one another:

$$\begin{aligned} ij &= k, & jk &= i, & ki &= j, \\ ji &= -k, & kj &= -i, & ik &= -j. \end{aligned}$$

A convenient way to remember the pattern is to place i, j, k around a circle in that order: multiplying two neighbours *in* the cyclic direction gives the third with a plus sign, and multiplying them *against* the cyclic direction gives the third with a minus sign.

$\times$	1	i	j	k
1	1	i	j	k
i	i	-1	k	$-j$
j	j	$-k$	-1	i
k	k	j	$-i$	-1

Read the table as (row) $\times$ (column); the order matters, and that is the whole point.

Remark (Quaternion multiplication is not commutative). Look at the table entries for ij and ji : one is k and the other is $-k$. So $ij \neq ji$. This is a genuine break with every number system you have used before — in $\mathbb{R}$ and in $\mathbb{C}$, xy and yx are always the same. In $\mathbb{H}$ they need not be, and swapping the order of a product can change the answer's sign. Hamilton's insight was that this sacrifice is unavoidable: you cannot have a four-dimensional number system with division *and* commutativity.

10.3. Computing with quaternions

Addition and subtraction are componentwise, just as in $\mathbb{C}$ — add the four coefficients slot by slot:

$$(a_1 + b_1i + c_1j + d_1k) + (a_2 + b_2i + c_2j + d_2k) = (a_1 + a_2) + (b_1 + b_2)i + (c_1 + c_2)j + (d_1 + d_2)k.$$

Multiplication is done exactly as in $\mathbb{C}$: distribute every term against every term, then reduce each product of basis elements using the table. The only new discipline is that you must preserve the order of the factors while distributing.

Example 12 (A product, and the same product reversed). Let $q = 1 + i$ and $p = 1 + j$. Then

$$qp = (1 + i)(1 + j) = 1 + j + i + ij = 1 + i + j + k,$$

using $ij = k$. Now reverse the order:

$$pq = (1 + j)(1 + i) = 1 + i + j + ji = 1 + i + j - k,$$

using $ji = -k$. The two answers differ in the sign of the k term: $qp \neq pq$. Noncommutativity is not an exotic edge case — it shows up in the simplest example available.

Conjugate and modulus generalise directly from $\mathbb{C}$. The conjugate flips the sign of every imaginary component,

$$\bar{q} = a - bi - cj - dk,$$

and the modulus is the four-dimensional Pythagoras,

$$|q| = \sqrt{a^2 + b^2 + c^2 + d^2}, \quad \text{with} \quad q\bar{q} = |q|^2.$$

A quaternion with $|q| = 1$ is called a **unit quaternion**, and just as the unit complex numbers formed a circle, the unit quaternions form a sphere — a three-dimensional one, sitting in four-dimensional space.

Division works as in $\mathbb{C}$, and for the same reason: since $q\bar{q} = |q|^2$ is a positive real number whenever $q \neq 0$, every nonzero quaternion has an inverse, namely $q^{-1} = \bar{q}/|q|^2$. So $\mathbb{H}$ supports all four arithmetic operations. It is a *division algebra* — a field in every respect except commutativity.

10.4. What they are used for

Unit quaternions do for rotation in three dimensions what unit complex numbers did for rotation in the plane in §8. A single unit quaternion encodes a rotation about an arbitrary axis in space; composing rotations is quaternion multiplication, and undoing one is conjugation. This is not a theoretical observation — it is how rotation is actually implemented in:

- **Computer graphics and game engines**, for orienting cameras, characters and objects, and for smoothly interpolating between two orientations;
- **Robotics**, for representing the pose of a joint, a limb, or a whole vehicle;
- **Aerospace**, for spacecraft and aircraft attitude control, where an orientation must be tracked continuously and reliably;
- **Sensor fusion**, for combining accelerometer, gyroscope and magnetometer readings into a single orientation estimate — the computation your phone performs to know which way it is being held.

Three practical advantages explain the popularity. Quaternions are *compact*: four numbers instead of the nine in a rotation matrix. They are *numerically well behaved*: accumulated rounding error is corrected simply by rescaling to unit modulus. And they avoid *gimbal lock*, the failure mode in which a rotation described by three separate angles loses a degree of freedom in certain configurations and the description breaks down. For anything that must rotate freely and continuously, quaternions are the standard tool.

Optional material

Neither appendix is required. The first is here because the derivation is genuinely beautiful and several of you will want it; the second collects facts about quaternions that did not fit the main thread.

A. Optional: deriving Euler's formula

Euler's formula was asserted in §5.2. Here is where it comes from. The only input is three Maclaurin series you have met in calculus.

The three series

Each of these converges for every real x :

$$\begin{aligned}e^x &= \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \cdots \\ \cos x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots \\ \sin x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots\end{aligned}$$

Notice that between them, $\cos$ and $\sin$ use every power of x exactly once: cosine takes the even powers, sine takes the odd ones. That is the observation the whole derivation turns on.

The powers of i cycle

We will need i^n . Because $i^2 = -1$,

$$i^0 = 1, \quad i^1 = i, \quad i^2 = -1, \quad i^3 = i^2 \cdot i = -i, \quad i^4 = (i^2)^2 = 1,$$

and from there it repeats with period 4: $1, i, -1, -i, 1, i, \dots$. Note the pattern of the two rows: at even powers i^n is real and alternates in sign, and at odd powers it is imaginary and alternates in sign.

The substitution

Feed ix into the exponential series in place of x :

$$e^{ix} = \sum_{n=0}^{\infty} \frac{(ix)^n}{n!} = 1 + ix + \frac{(ix)^2}{2!} + \frac{(ix)^3}{3!} + \frac{(ix)^4}{4!} + \frac{(ix)^5}{5!} + \frac{(ix)^6}{6!} + \cdots$$

Each term splits as $(ix)^n = i^n x^n$, so apply the cycle above:

$$e^{ix} = 1 + ix - \frac{x^2}{2!} - i\frac{x^3}{3!} + \frac{x^4}{4!} + i\frac{x^5}{5!} - \frac{x^6}{6!} - \cdots$$

Separating the two kinds of term

Now sort the terms into those without i and those with it. The even powers of x carry no i ; the odd powers all do:

$$e^{ix} = \underbrace{\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots\right)}_{\cos x} + i \underbrace{\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots\right)}_{\sin x}.$$

Both bracketed series are on the list we started with. Therefore

$$e^{ix} = \cos x + i \sin x,$$

which is Euler's formula. ■

Remark. One step deserves a footnote: rearranging infinitely many terms into two groups is not automatically legal. It is legal here because the exponential series converges *absolutely*, and absolutely convergent series may be rearranged freely without changing their sum. That fact is proved in a course on analysis.

Remark. The derivation also explains why the formula *had* to look like this. The exponential's series uses every power of x ; cosine claims the even ones and sine the odd ones; and multiplying by i is exactly what flips a term between the two families. Euler's formula is the bookkeeping identity that results. Setting $x = \pi$ then gives $e^{i\pi} = -1$, and Euler's identity is a corollary of a rearrangement.

B. Optional: more on quaternions

The general product

Multiplying out $q_1 q_2$ in full, with $q_1 = a_1 + b_1 i + c_1 j + d_1 k$ and $q_2 = a_2 + b_2 i + c_2 j + d_2 k$, and reducing every basis product by the table of §9.2, gives the *Hamilton product*:

$$\begin{aligned} q_1 q_2 = & (a_1 a_2 - b_1 b_2 - c_1 c_2 - d_1 d_2) \\ & + (a_1 b_2 + b_1 a_2 + c_1 d_2 - d_1 c_2) i \\ & + (a_1 c_2 - b_1 d_2 + c_1 a_2 + d_1 b_2) j \\ & + (a_1 d_2 + b_1 c_2 - c_1 b_2 + d_1 a_2) k. \end{aligned}$$

You are not expected to memorise this. It is worth seeing once, because it is what a graphics library actually executes, and because the asymmetry of the signs is where noncommutativity lives: swapping the roles of q_1 and q_2 flips the sign of some terms but not others.

Rotating a vector

To rotate a three-dimensional vector $\mathbf{v} = [v_1, v_2, v_3]$, embed it as a quaternion with zero scalar part, $v = v_1 i + v_2 j + v_3 k$, and conjugate by a unit quaternion q :

$$v \longmapsto q v q^{-1}.$$

If $q = \cos(\theta/2) + \sin(\theta/2) \mathbf{n}$ for a unit vector $\mathbf{n}$ written in i, j, k components, the result is $\mathbf{v}$ rotated by the angle θ about the axis $\mathbf{n}$. The half-angle is not a typo: q appears twice in the formula, once on each side, so each contributes half the rotation.

Why q and $-q$ give the same rotation

Replacing q by $-q$ leaves qvq^{-1} unchanged, because the two sign changes cancel. So each rotation of space corresponds to *two* unit quaternions. In the language of a later course, the unit quaternions form a *double cover* of the group of rotations of three-dimensional space — a fact with real consequences for interpolation, since a naive average of two orientations can travel the long way around unless the signs are chosen consistently.

Why the sequence stops

The pattern $\mathbb{R}$ (dimension 1), $\mathbb{C}$ (dimension 2), $\mathbb{H}$ (dimension 4) invites a guess about dimension 8, and there is such a system: the *octonions* $\mathbb{O}$. But each extension costs a property. Passing from $\mathbb{R}$ to $\mathbb{C}$ costs the ordering — there is no sensible way to say one complex number is less than another. Passing from $\mathbb{C}$ to $\mathbb{H}$ costs commutativity. Passing from $\mathbb{H}$ to $\mathbb{O}$ costs associativity, so that $(xy)z$ and $x(yz)$ can differ. And there the sequence ends: a theorem of Frobenius says that $\mathbb{R}$, $\mathbb{C}$ and $\mathbb{H}$ are the *only* finite-dimensional associative division algebras over the reals. Hamilton's four dimensions were not a lucky guess; they were the last stop before the structure gives way.